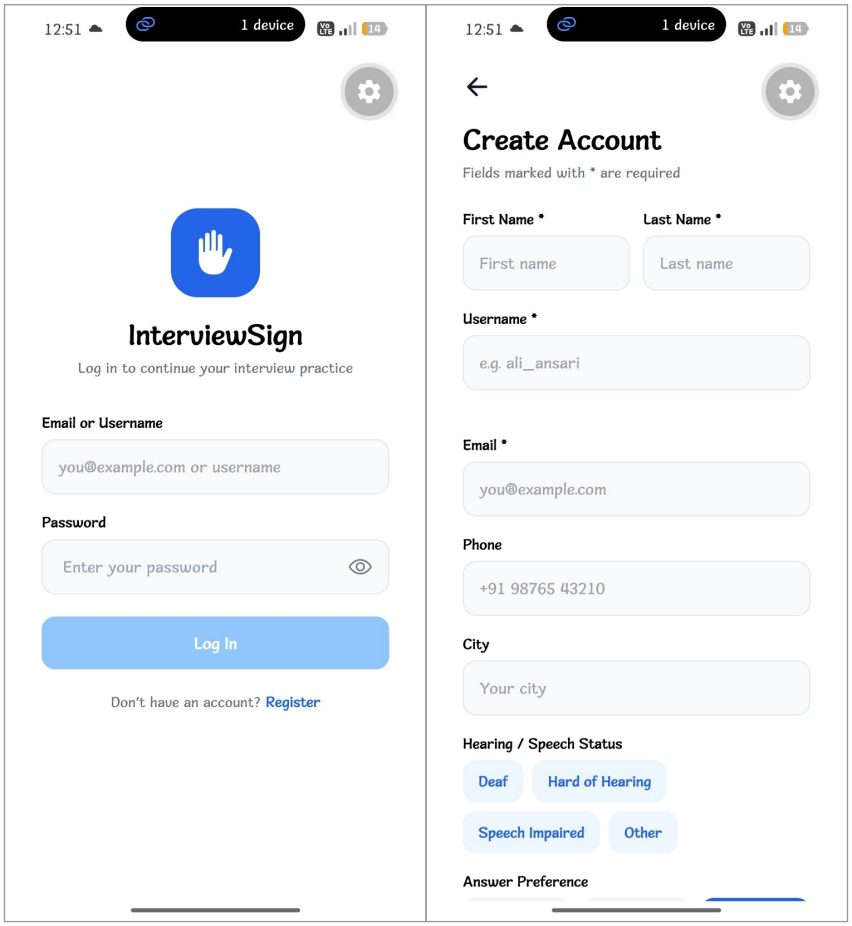
CHAPTER 5
RESULTS AND DISCUSSIONS

5.1 Authentication Screen

The Authentication screen is the first screen of InterviewSign. It is shown only to users who are not logged in, because the protected navigation hides the other screens until a session exists. Fig. 5.1 shows the Login and Register screens.



Login Screen *Register Screen*

Fig. 5.1: Login and Register Screen

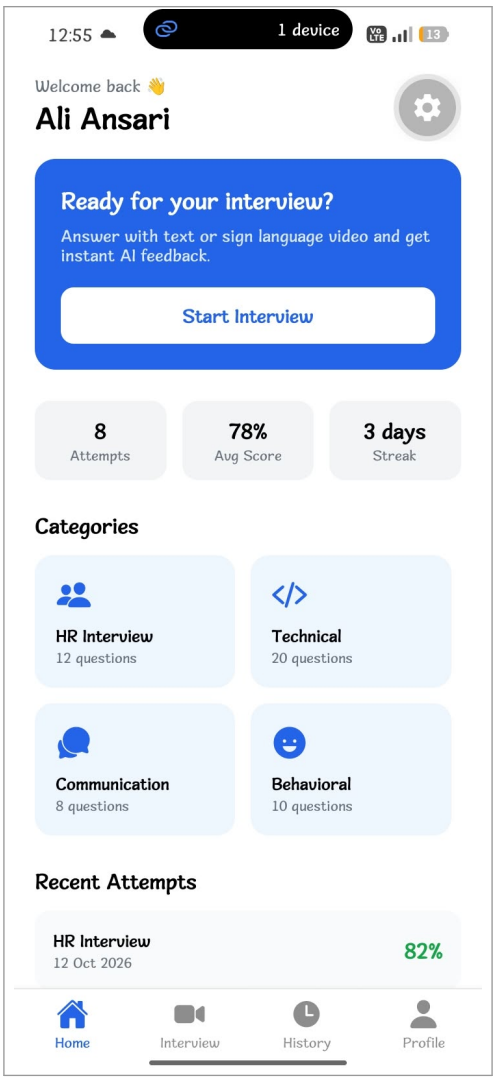
Login. The user types an email address or a username in one field and the password in the other; an eye icon shows or hides the password. The Log In button stays light blue and disabled until the identifier is filled and the password has at least 6 characters. If a username is entered, the application finds the matching email through the Supabase function get_email_by_username and then signs in with Supabase Authentication. A wrong entry gives one general message, "Invalid email/username or password.", which does not reveal whether the username or the password was wrong.

Register. The Create Account form marks the required fields with *. It asks for first name, last name, username, email, phone, city, hearing or speech status (Deaf, Hard of Hearing, Speech Impaired or Other) and answer preference (Text, Sign Video or Both), followed by the password and its confirmation. The username field accepts only lowercase letters, numbers and underscore, and after a 500 ms pause the application shows whether the name is available. The Register button becomes active only when all required fields are valid. On success, the profile is created automatically from the entered details and the user is logged in.

Discussion. Login with either an email or a username, the live username check and the large, high-contrast fields keep the first step simple and accessible. Optional details such as phone and city can be left empty.

5.2 Home Screen

The Home screen opens after a successful login. It is the first of the four tabs (Home, Interview, History and Profile) at the bottom of the application and gives a quick summary of the user's practice. Fig. 5.2 shows the Home screen.



Home Screen

Fig. 5.2: Home Screen

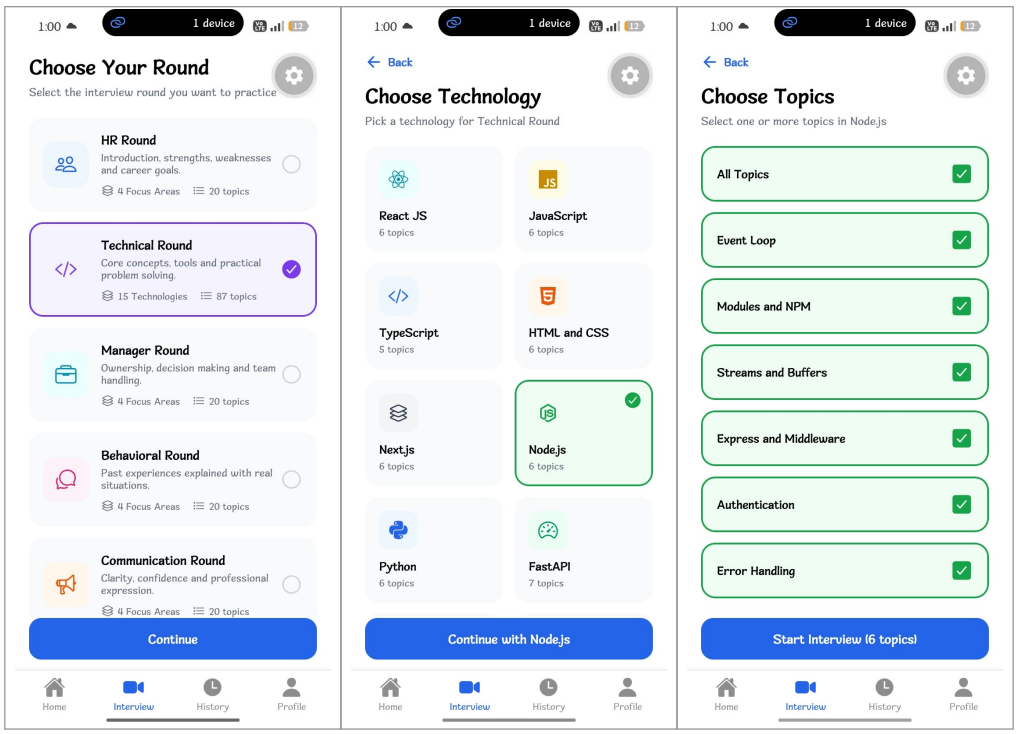
Description. The top of the screen greets the user by name. The blue Start Interview card invites the user to answer with text or sign language video and opens the Interview tab. Below it, three boxes show the number of Attempts, the Average Score and the Streak in days. The Categories grid lists HR Interview (12 questions), Technical (20), Communication (8) and Behavioral (10). The Recent Attempts list shows the title, date and score of the latest attempts, and the tab bar at the bottom moves between the main screens.

Result. Tapping a category card opens the History screen with that category already selected, so the user can see earlier attempts of that type at once. Scores are shown in colour (green for 75 and above), which makes progress easy to read without reading each number.

Discussion. Showing the greeting, the main action, the statistics and the categories on one screen means the user reaches any activity in one tap. The large cards, clear icons and simple words support Deaf, Hard of Hearing and Speech Impaired users.

5.3 Interview Screen

The Interview screen is the second tab of the bottom bar. It opens when the user taps the Start Interview card on the Home screen or the Interview tab. The screen works as a three-step wizard in which the user selects the interview round, then the technology or focus area, and then the topics to practise. Fig. 5.3 shows the three steps.



Choose Round *Choose Technology* *Choose Topics*

Fig. 5.3: Interview Screen

Description. *Step 1, Choose Your Round:* The first step lists five rounds as large cards: HR Round, Technical Round, Manager Round, Behavioral Round and Communication Round. Each card shows an icon, a one-line description, the number of focus areas or technologies, and the total number of topics. For example, the Technical Round offers 15 technologies and 87 topics, while the other rounds offer 4 focus areas and 20 topics each. The chosen card gets a coloured border and a tick, and the blue Continue button moves to the next step.

Step 2, Choose Technology: The second step shows the options of the selected round in a two-column grid. For the Technical Round the options are technologies such as React JS, JavaScript, TypeScript, HTML and CSS, Next.js, Node.js, Python and FastAPI, each with its icon and number of topics. The heading of this step changes with the round. The selected card is highlighted in green and the button text changes to "Continue with Node.js". A Back link at the top returns to the previous step.

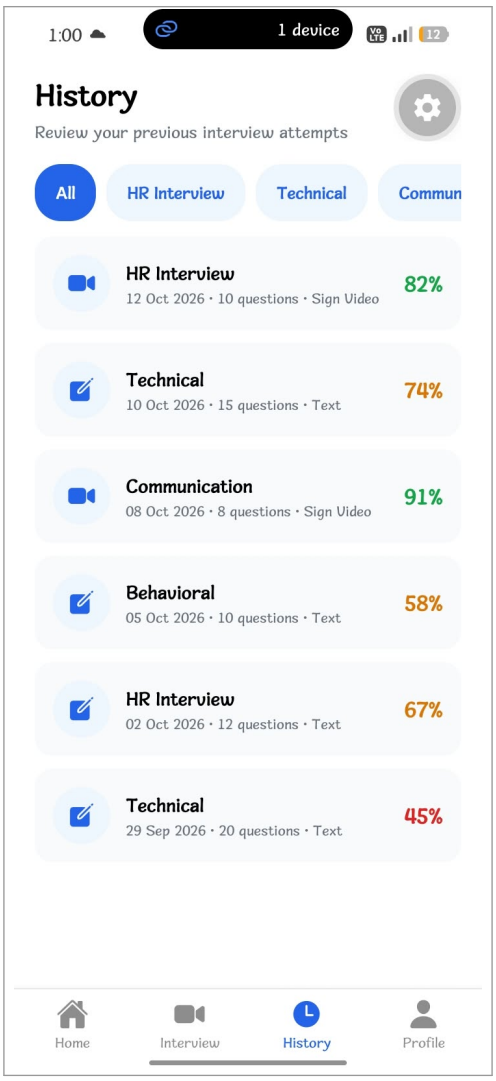
Step 3, Choose Topics: The third step shows the topics of the chosen technology as a checklist. For Node.js the topics are Event Loop, Modules and NPM, Streams and Buffers, Express and Middleware, Authentication and Error Handling. The All Topics option at the top selects or clears every topic at once. The final button shows the number of selected topics, for example "Start Interview (6 topics)".

Result. The steps open in order, and the Back button clears the later choices, so the user never sees an old selection after changing the round. Changing the round also removes the earlier option and topics. The final button stays disabled with the text "Select at least one topic" until at least one topic is ticked, which prevents an empty interview from starting. The accent colour changes with the selected item (purple for the round, green for the technology and topics), which shows the user clearly what has been chosen.

Discussion. Dividing the selection into three short steps keeps each screen simple, with only one decision at a time. The large cards, clear icons, short descriptions and visible topic counts let the user choose without reading long text, which supports Deaf, Hard of Hearing and Speech Impaired users. Because the rounds, options and topics come from a common data structure, new rounds or technologies can be added without changing the screens.

5.4 History Screen

The History screen is the third tab of the bottom bar. It keeps the record of all earlier interview attempts of the logged-in user, so that the user can review the practice and see the progress over time. It opens from the History tab or from a category card on the Home screen. Fig. 5.4 shows the History screen.



History Screen

Fig. 5.4: History Screen

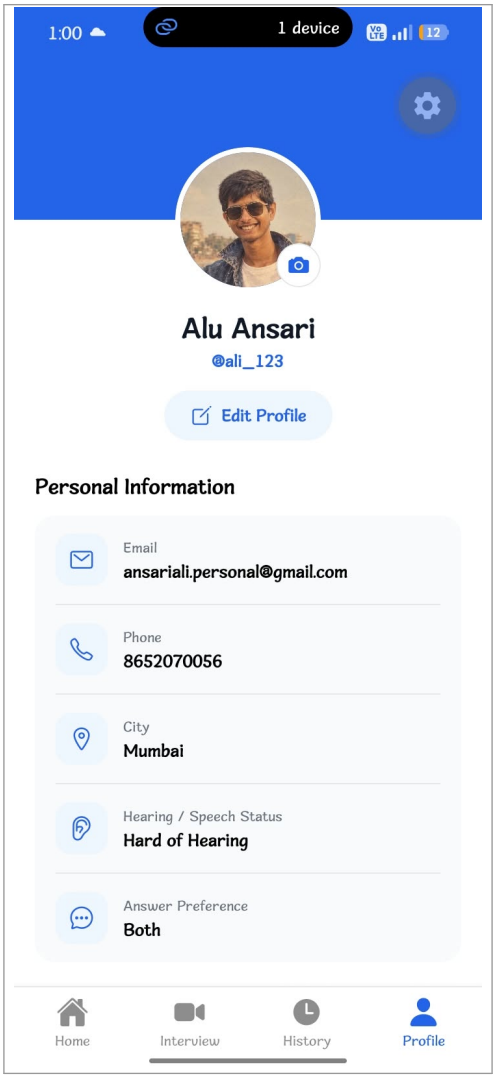
Description. The top of the screen shows the title History and the line "Review your previous interview attempts". Below it, a row of filter chips (All, HR Interview, Technical, Communication and Behavioral) can be scrolled sideways. The selected chip is filled with blue. Each attempt is shown as a card with an icon, the category, the date, the number of questions, the answer type (Text or Sign Video) and the score in percentage. A video camera icon marks an attempt answered by sign video and a pencil icon marks an attempt answered by text.

Result. Selecting a chip shows only the attempts of that category, and the chip moves to the centre of the row. When the user comes from a category card on the Home screen, the matching chip is already selected. The score is shown in colour: green for 75 and above (82% and 91%), amber for 50 to 74 (74%, 58% and 67%) and red for below 50 (45%). When no attempt matches the selected chip, the message "No attempts found" is displayed instead of an empty list.

Discussion. The History screen lets the user compare attempts by category, answer type and score at a glance. The colour of the score and the icon of the answer type give the information without reading long text, which is helpful for Deaf, Hard of Hearing and Speech Impaired users. The attempts are stored for each user in Supabase, so every user sees only personal records.

5.5 Profile Screen

The Profile screen is the fourth and last tab of the bottom bar. It shows the personal details that the user entered at registration and lets the user change the profile photograph or open the Edit Profile screen. Fig. 5.5 shows the Profile screen.



Profile Screen

Fig. 5.5: Profile Screen

Description. The screen starts with a blue header banner and a large circular profile photograph. If the user has not added a photograph, the initials of the name are shown instead. A small camera button on the photograph opens the gallery. Below the photograph, the full name, the username with the @ sign and the Edit Profile button are displayed. The Personal Information card lists Email, Phone, City, Hearing / Speech Status and Answer Preference, each with its own icon. A field that the user left empty shows the text "Not added". The Logout button is placed at the bottom of the screen.

Result. When the user selects a new photograph from the gallery, it is cropped to a square, uploaded to the avatars bucket of Supabase Storage and its public link is saved in the profile. The new photograph appears in the header at once. The Edit Profile button opens the Edit Profile screen, where the details and the username can be changed, and the username can be changed only once in 7 days. The Logout button ends the session and returns the user to the Login screen.

Discussion. Keeping all personal details on one screen with clear icons and large text makes the profile easy to read and update. The Hearing / Speech Status and Answer Preference fields show the accessibility choices of the user, so the application can be used in the way that suits the user best. The email is shown for reading only, which protects the account identity.